

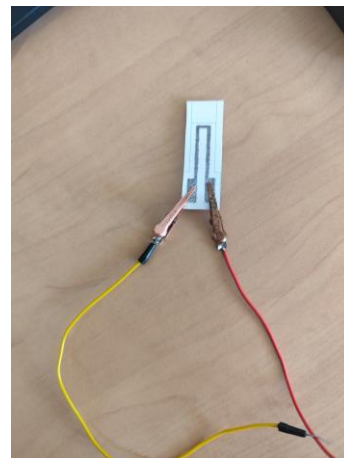
General features

- Low power consumption
- Easy to use
- Low cost
- Detection of strain
- Different type of pen hardness

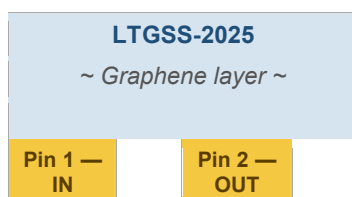
Description

This sensor is a strain gauge based on Graphene. Using a pencil, a layer of graphite is deposited on a sheet of paper and by exerting stress on the sensor, the Graphene lattice will be more or less spaced, and the electrons will flow more or less easily.

By using an amplifier, we can measure the signal modified by this stress and thus determine its value. The advantage of this sensor is that it is easy to use, can be modified to suit the type of pencil, and gives results comparable to those of industrial sensors.



Pin description



Pin number	Usage
1	In
2	Out

Specifications

Parameter	Value
Type	Graphene based sensor
Material	Graphene (from pencil)
Sensor type	Passive
Dimensions	3 x 1.5 cm ² — Thickness : 0.2 mm
Temperature range	10 to 35 °C
Pencil tone	H to 3B
Supply voltage	Maximum 5V (for Arduino)

Electrical characteristics

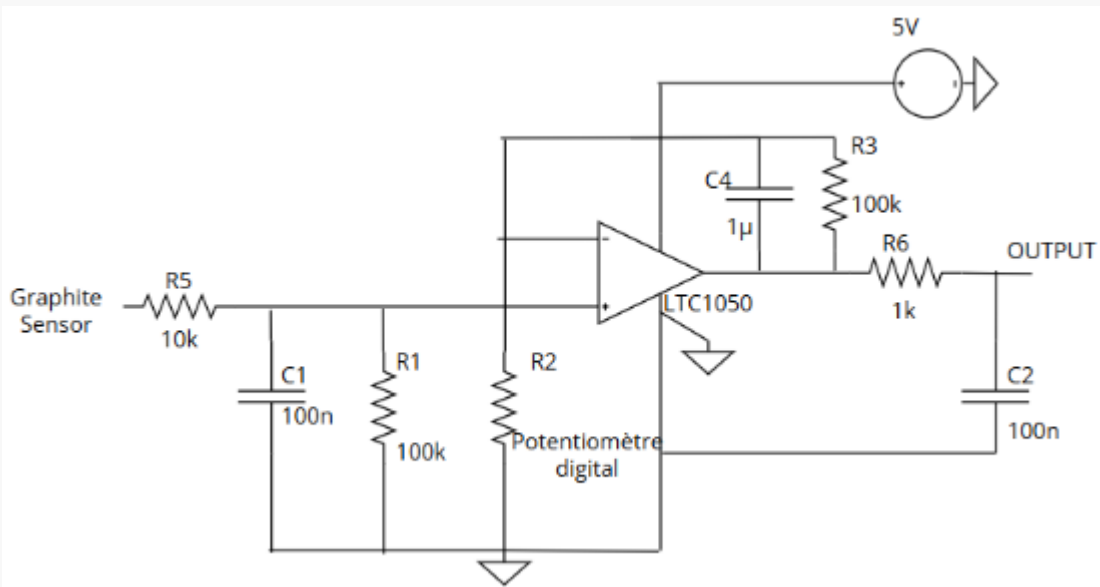
Resistance values in MΩ for different pencil tones.

Pencil tone	Units	Values		
		Min (compression)	Typ	Max (flexion)
4B	MΩ		140	
3B	MΩ	290	350	550
2B	MΩ	240	280	350
B	MΩ	300	340	480

Application

This is the recommended circuit to use with the LTGSS-2025. The LTC1050 is a chopper-stabilised op-amp used as a transimpedance amplifier. The digital potentiometer (MCP41010) sets the gain via R2.

Application circuit — LTGSS-2025



Thanks to this circuit, it is possible to calculate the resistance of the graphite sensor using the following equation:

$$R_{sensor} = \left(1 + \frac{R3}{R2}\right) R1 \cdot \frac{V_{cc}}{V_{adc}} - R1 - R5$$

Component values:

- R1 = 100 kΩ (reference resistor)
- R2 = digital potentiometer MCP41010 (0 to 10 kΩ)
- R3 = 100 kΩ (feedback resistor)
- R5 = 10 kΩ (series input resistor)
- R6 = 1 kΩ (output resistor)
- C1 = C2 = 100 nF (decoupling capacitors)
- C4 = 1 μF (supply decoupling)
- Vcc = 5 V (Arduino supply voltage)
- Vadc = voltage measured on ADC pin

Sensibiliter en fonction du type de crayon

